

EE Dept, IIT Bombay
Academic Year: 2025-2026, Semester I (Autumn)
Course: MS101 Makerspace
EE Lecture 1: Devices and Circuits I
Sample Problems

1. (A) Show that a series connection of two resistances R_1 and R_2 is equivalent to a single resistance R_e given as

$$R_e = R_1 + R_2$$

- (B) Show that a parallel connection of two resistances R_1 and R_2 is equivalent to a single resistance R_e given as

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2}$$

Hint: (A) Apply a current i through the series connection. Find the total voltage in terms of voltages across the individual resistances. (B) Apply a voltage v across the parallel connection. Find the total current in terms of currents in the individual resistances.

2. (A) Find the resistance to be connected in series with $100\text{ k}\Omega$ to get $110\text{ k}\Omega$.
(B) Find the resistance to be connected in parallel with $100\text{ k}\Omega$ to get $90\text{ k}\Omega$.
3. (A) Show that a parallel connection of two capacitances C_1 and C_2 is equivalent to a single capacitance C_e given as

$$C_e = C_1 + C_2$$

- (B) Show that a series connection of two capacitances C_1 and C_2 is equivalent to a single capacitance C_e given as

$$\frac{1}{C_e} = \frac{1}{C_1} + \frac{1}{C_2}$$

Hint: Same as for Q, 1

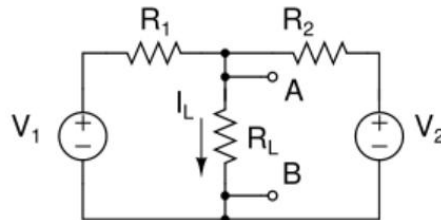
4. (A) Find the capacitance to be connected in parallel with 100 nF to get 120 nF .
(B) Find the capacitance to be connected in series with 100 nF to get 80 nF .
5. A voltage attenuator is formed by applying the input voltage V_i across the series connected resistances R_1 and R_2 and taking the output voltage V_o across R_2 . Find the attenuation factor $a = V_o/V_i$. If R_2 is $2\text{ k}\Omega$, find R_1 for attenuation factor of $1/10$.
6. Mark all correct statements.
- (A) All capacitors have polarity marks on their terminals.
 - (B) Some resistors have polarity marks on their terminals.
 - (C) Inductors do not have polarity marks on their terminals.
 - (D) An ideal voltage source has zero source resistance.
 - (E) An ideal current source has zero source resistance.

7. A transformer may be damaged if the voltage applied on its primary or secondary side exceeds the corresponding rated voltage. A transformer is rated as 230 V to 12 V.
- (A) What is the primary-to-secondary turns ratio of this transformer?
- (B) What will happen when 10 V is connected to the primary side?
- (C) What will happen when 6 V is connected to the secondary side?
- (D) What will happen when 400 V is connected to the primary side?
- (E) What will happen when 24 V is connected to the secondary side?

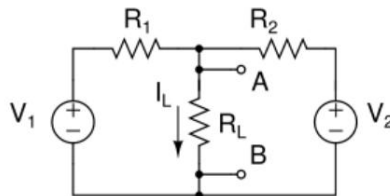
8. Optional problem: A transformer with primary-to-secondary turns ratio of $a : 1$ is connected to a voltage V_1 on the primary side and load R_L on the secondary side.
- (A) Find the secondary side output voltage V_2 .
- (B) Find the load current I_2 .
- (C) Find the current I_1 drawn from the input voltage source. Show that this current is the same if V_1 is connected across a resistance $R_e = a^2 R_L$.

Note: Most visible application of transformers is as step-up and step-down transformers for increasing and decreasing the voltage in power applications. They are also used for electrically isolating the secondary side from the primary side, and these transformers are known as isolation transformers. In a transformer with turns ratio of $a : 1$, a resistance R_L connected to the secondary side appears as a resistance $R_e = a^2 R_L$ connected across the input voltage. Based on this property, transformers are used as impedance converters in many electronic circuits.

9. In the circuit shown below, $V_1 = 10$ V, $V_2 = 15$ V, $R_1 = 2$ k Ω and $R_2 = 3$ k Ω . Calculate the current I_L in mA for $R_L = 2.8$ k Ω and $R_L = 1.3$ k Ω .

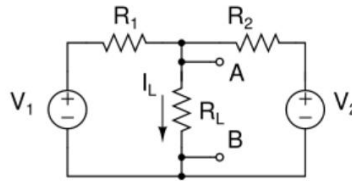


10. In the circuit shown below, $R_1 = R_2 = R_L = 1200$ Ω .
- (A) If $V_1 = 9$ V, and $V_2 = 0$ V, calculate V_{AB} in volts and I_L in mA.
- (B) If $V_1 = 0$ V, and $V_2 = 12$ V, calculate V_{AB} in volts and I_L in mA.
- (C) If $V_1 = 9$ V, and $V_2 = 12$ V, calculate I_L in mA.

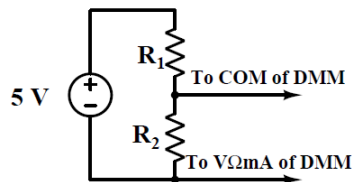


11. A practical voltage source has an open circuit voltage of 10 V. When a 1.75 k Ω resistor is connected as the load across its terminals, the terminal voltage drops to 8.75 V. What is the source resistance in ohms?

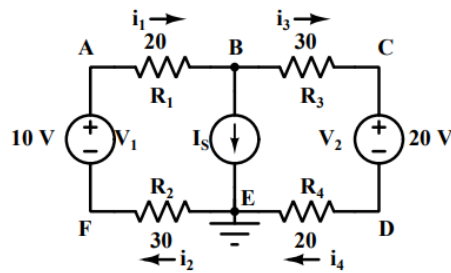
12. A practical current source is modelled as an ideal current source of 100 mA in parallel with a resistor of 240 k Ω . A load resistance of 10 k Ω is connected as the load across its terminals. What is the load current?
13. A resistive network is shown below. Given: $R_1 = R_2 = R_L = 3000\ \Omega$. $V_1 = -15\text{ V}$, and $V_2 = 6\text{ V}$. Using superposition method, calculate V_{AB} and I_L .



14. In the circuit shown below, two resistors R_1 and R_2 are connected to a 5 V DC source. Given: $R_1 = R_2 = 2\text{ M}\Omega$. Note: $1\text{ M}\Omega = 10^6\ \Omega$. The voltage across R_2 is measured using a DMM by putting it in the DC Voltage mode. The DMM in the chosen voltage range can show two decimal places and has an internal resistance of 10 M Ω . What reading will you see on the DMM? Note: The DMM's input resistance is in parallel with R_2 .

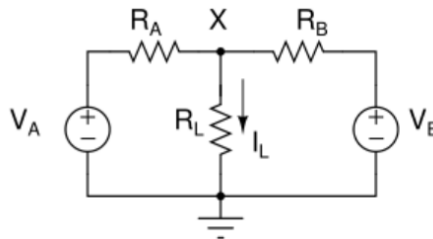


15. A voltage source is made by connecting two 12 V batteries in series, each having an internal resistance of 5 Ω .
- (A) What's the no-load voltage of this voltage source? No-load voltage is the voltage across the terminals of a voltage source when the load resistance is infinite.
- (B) A load resistance of 100 Ω is connected across the terminals of this voltage source. What will be the terminal voltage?
- (C) How much power in watts will be dissipated in the 100 Ω load resistor?
16. A practical current source is modelled as an ideal current source of 200 mA in parallel with a resistance of 150 k Ω . A load resistor R_L is connected across this source. The current through R_L is measured as 125 mA. What is R_L in k Ω ?
17. In the circuit shown below, positive current directions are shown by arrow. Given: $V_1 = 10\text{ V}$, $V_2 = 20\text{ V}$, $I_S = 0.5\text{ A}$. $R_1 = 20\ \Omega$, $R_2 = 30\ \Omega$, $R_3 = 30\ \Omega$, $R_4 = 20\ \Omega$.
- (A) Find i_1 , i_2 , i_3 , and i_4 .
- (B) Find v_A , v_B , and v_C (voltages at nodes A, B, and C with respect to node E).



18. A practical current source is modelled as an ideal current source I_S in parallel with a resistance of $80\text{ k}\Omega$. When a resistance of $4\text{ k}\Omega$ is connected across its terminals, the voltage is found to be 20 V . What is I_S in mA?

19. Using the superposition method, find the node voltage V_X and the current I_L in the circuit shown below, given: $V_A = 18\text{ V}$, $V_B = -9\text{ V}$, $R_A = R_B = R_L = 600\text{ }\Omega$.



20. A practical voltage source is modelled as an ideal voltage source of 15 V in series with a source resistance R_S . When a load resistance is connected across this source, the current in the load is 400 mA and the voltage across it is 13 V .
- (A) Evaluate R_S in ohms.
- (B) What is the load voltage if the load resistance is $400\text{ }\Omega$?

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